

SARAVANAMUTHU S

AI & Agentic Systems Engineer — LLM Platforms • Backend • Distributed Systems

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PROFESSIONAL SUMMARY

AI engineer specializing in production agentic systems and LLM platforms, with 3+ years building the backend and orchestration layer of an autonomous multi-agent product. Promoted to Senior Software Engineer within two years and now lead a team of 6–8 engineers. Deep hands-on work across multi-agent orchestration, RAG and vector retrieval, context engineering, and self-improving learning systems, on a strong Go and distributed-systems foundation.

TECHNICAL SKILLS

AI & LLMs: LLM Integration & Orchestration, Multi-Agent / Agentic Systems, RAG & Vector Retrieval, Embeddings, Context Engineering, Prompt Engineering, Model Context Protocol (MCP), Event-Sourced Learning Systems

Languages: Go (Golang), Node.js / JavaScript, Python

Backend & Real-Time: REST APIs, Microservices, WebSockets, Server-Sent Events (SSE), Kafka, DAG Orchestration, Event Sourcing

Data & Vector: MongoDB, Redis, PostgreSQL, Milvus / Zilliz (Vector DB), ETL Pipelines

Cloud & DevOps: Azure (Entra ID, VM, AKS, Container Apps, Container Registry, AI Foundry), AWS (Bedrock, IAM, EC2), GCP (Compute Engine), Docker, Jenkins, Nginx

Concepts: Distributed Systems, Real-Time Architectures, System Design, Performance Optimization, Concurrency, Security Testing

WORK EXPERIENCE

Adya.ai Aug 2023 – Present

Senior Software Engineer Aug 2025 – Present

- Lead a team of 6–8 engineers, owning system design, architectural standards, and code quality across backend services.
- Own and evolve the autonomous multi-agent platform (Super AI) — turning plain-language requests into delivered artifacts through a DAG engine (dependency resolution, retries, cascade-failure handling) that runs ~12 specialist sub-agents in parallel or sequence.
- Drive development of the AI Agent Builder — an n8n-style natural-language-to-workflow system — built on a master-LLM recursive tool loop grounded in a 750+ component catalog via vector retrieval.
- Designed Auto Compact, a 3-layer context-management system (cross-request, per-call, and per-tool-call compaction) that keeps long agent-building sessions within LLM token limits, and architected long-running compute for persistent multi-step execution.
- Architected a self-improving, event-sourced learning layer that extracts durable knowledge from agent decisions and injects it into future runs under a token budget — with no gradient training — improving workflow synthesis over time.
- Led black-box security testing with Strix (an open-source AI pen-testing agent), triaging and remediating 5 critical and 7 high-severity vulnerabilities ahead of release.

Software Developer Aug 2023 – Jul 2025

- Optimized the production backend — MongoDB connection pooling and reuse, missing-index remediation, and Redis caching — cutting RAM usage ~30% under peak load; absorbed burst traffic with Go routines and channels at stable latency.
- Designed a central skill registry — metadata indexed in a vector database, raw definitions in MongoDB — equipping 1,400+ open-source skills on demand at runtime, auto-installing new capabilities from an online marketplace when no local match exists.
- Built real-time collaboration and streaming with WebSockets and Server-Sent Events for live multi-user editing and low-latency updates.
- Architected Cloud Studio, a self-service PaaS/SaaS deployment platform that integrates the DevOps team's infrastructure-provisioning APIs — provisioning servers and deploying apps and agents across Azure (VM/AKS), GCP, and AWS (Bedrock), with Azure AI Foundry for model and key management; shipped via Docker and Jenkins.
- Built no-code ETL pipelines across databases, cloud storage, and SaaS sources, and enabled AI browser automation for agent-driven workflows.

Eunimart Aug 2022 – Jul 2023

Software Engineer Intern Aug 2022 – Jul 2023

- Engineered ONDC-compliant backend services for a centralized e-commerce platform in Go and Node.js, and built the ONEST platform delivering course and job provider–seeker marketplaces on a unified backend.
- Developed an ONDC Reconciliation Service Provider (RSP) automating transaction reconciliation and settlement, integrating banking partners HSBC, IDFC, and NPCI into secure automated pipelines.
- Integrated marketplaces, webstores, and logistics providers for omnichannel seller operations, owning production backend services end-to-end.

EDUCATION

B.E. Information Science & Engineering — CGPA 8.9 *Graduated 2023*

Bannari Amman Institute of Technology, Sathyamangalam